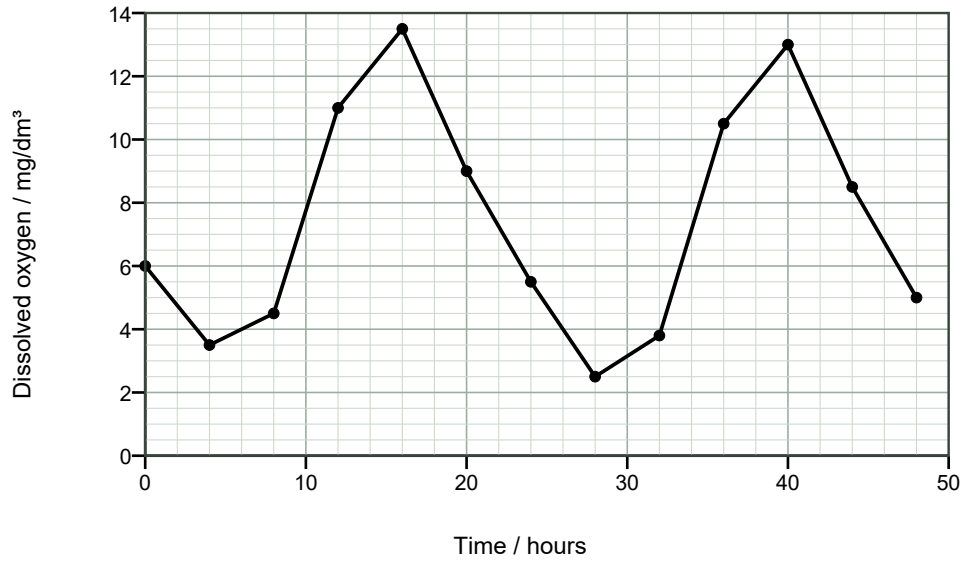
Questions by topic: Nutrient pollution, sewage treatment & acid mine drainage - Set 1 (70 marks)

Answer all questions in the spaces provided.

0 1 Figure 1 shows the dissolved oxygen concentration in a shallow lowland lake over 48 hours in late summer. The lake receives runoff from arable farmland and effluent from a sewage works.

Figure 1



0 1 . 1 Describe the pattern shown in Figure 1.

[2 marks]

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0 1 . 2 Explain the causes of the pattern shown in Figure 1.

[4 marks]

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0 1 . 3 Explain why fish kills in eutrophic lakes usually happen at night or at the end of summer.

[2 marks]

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0 1 . 4 Give **two** other harmful effects of algal blooms.

[2 marks]

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0 2 . 1 Explain why nitrates in drinking water can cause **blue baby syndrome** (methaemoglobinaemia).

[3 marks]

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0 2 . 2 Explain why the link between nitrates in drinking water and stomach cancer is uncertain.

[2 marks]

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0 2 . 3 Describe **three** ways farmers in a Nitrate Vulnerable Zone can reduce nitrate leaching into aquifers.

[3 marks]

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0 2 . 4 Explain how phosphates are removed from sewage effluent and from lakes polluted in the past.

[2 marks]

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0 3 Figure 1 is a flow diagram of the main stages in a sewage treatment works using activated sludge treatment.

Figure 1

Stage	Process
Pre-treatment	Screens and grit traps
Primary treatment	Primary sedimentation
Secondary treatment	Aeration tank then secondary sedimentation
Tertiary treatment	Phosphate stripping / micro-strainers / UV
Sludge treatment	Anaerobic digestion

0 3 . 1 Explain the purpose of screens and grit traps.

[2 marks]

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0 3 . 2 Explain why primary sedimentation is described as the most important process.

[2 marks]

0 3 . 3 Describe what happens in the aeration tank and explain why the sludge from the secondary sedimentation tank is returned to it.

[3 marks]

0 3 . 4 Explain why tertiary treatment may be needed before effluent is discharged, and describe **one** tertiary process.

[3 marks]

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0 4 . 1 Describe the process of anaerobic sludge digestion and give **two** benefits of the process.

[4 marks]

0 4 . 2 Explain **two** problems of using sewage sludge as an agricultural fertiliser and how each can be reduced.

[4 marks]

0 4 . 3 Explain why it is more efficient to have separate collection systems for foul sewage and for surface drainage water.

[2 marks]

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0 5 Table 1 shows measurements taken from a stream above and below the point where drainage water from an old metal-mine spoil heap enters it.

Table 1

Site	pH	Dissolved zinc / mg dm^{-3}	Dissolved oxygen / % saturation	Number of invertebrate taxa	Mayfly larvae present?
Upstream	7.1	0.02	96	18	Yes
50 m downstream	3.4	6.5	94	3	No
2 km downstream	5.9	1.1	95	9	No

0 5 . 1 Explain how the acidic drainage water is produced in the spoil heap.

[3 marks]

0 5 . 2 Use Table 1 to explain why the number of invertebrate taxa falls so sharply 50 m downstream, even though dissolved oxygen is unchanged.

[3 marks]

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0 5 . 3 Suggest why the pH and zinc concentration have partly recovered 2 km downstream.

[2 marks]

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0 5 . 4 Describe how acid mine drainage can be controlled.

[2 marks]

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0 6 . 1 Explain why organic pollution of rivers is usually monitored by measuring Biochemical Oxygen Demand (BOD) rather than the concentrations of the pollutants themselves.

[3 marks]

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0 6 . 2 Explain why a coliform count is used to assess the health risk from sewage contamination rather than testing for pathogens such as typhoid.

[2 marks]

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0 6 . 3 Give **three** characteristics of species that make them useful in a biotic index.

[3 marks]

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0 6 . 4 Give **one** advantage and **one** disadvantage of using a biotic index rather than chemical monitoring.

[2 marks]

Advantage .....

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Disadvantage .....

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0 7 . 1 Explain how organic wastes and inorganic nutrients cause water pollution and evaluate the effectiveness of the methods used to control them.

[9 marks]

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0 7 . 2 Give **one** reason why algae cause eutrophication problems more quickly than rooted plants when nutrients are added to a lake.

[1 mark]

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END OF QUESTIONS

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